BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Programming Languages and Translators

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How many input rows are needed for three independent Boolean inputs?

2. What does software copyright restrict?

A) Using any keyboard
B) Learning to code
C) Copying without permission
D) Saving your own work

3. 32 items are packed in boxes of 5. Use DIV and MOD to give the number of full boxes and leftover items.

4. Entry needs a valid ticket AND membership card. "AND is true if just one input is true." Correct this. What happens with only a ticket?

Today's objectives

Explain the core ideas of programming languages and translators using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

High-level language

Human-readable abstraction; generally portable with suitable runtime/translator support.

Low-level language

Close to processor instruction architecture, giving greater hardware control.

Compiler

Translates source code into executable machine or intermediate output before running the resulting program.

Interpreter

Processes/translates and executes code using an interpreter during execution.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

High-level language	Human-readable abstraction; generally portable with suitable runtime/translator support.
Low-level language	Close to processor instruction architecture, giving greater hardware control.
Compiler	Translates source code into executable machine or intermediate output before running the resulting program.
Interpreter	Processes/translates and executes code using an interpreter during execution.

I Do - One worked example

A developer wants one program to run on different processors. Explain how language level and translators matter.

1. High-level code is less tied to a specific processor architecture.
2. A suitable translator/runtime must still exist for each target.
3. Low-level machine code is closely processor-specific.
4. Explain why a compiler/interpreter approach is chosen for the environment, not a guaranteed speed ranking.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Compare an interpreter and compiler in detecting errors.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the size calculation for four two-byte characters.

```
characters = 4
bytesPerCharacter = 2
sizeBytes = characters + bytesPerCharacter
print(sizeBytes)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Give one benefit of high-level languages. [2 marks]

Q3. Compare compiler and interpreter at a broad level. [4 marks]

Q4. Explain why machine code depends on processor architecture. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A high-level program still needs translation/runtime support to execute on a processor. ☐ T ☐ F
2. Human-readable abstraction; generally portable with suitable runtime/translator support. ☐ T ☐ F
3. Close to processor instruction architecture, giving greater hardware control. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 8, because 2 to the power 3 is 8; list every combination exactly once. ☐

Q2 C - Copying without permission ☐

Q3 32 DIV 5 = 6 full boxes; 32 MOD 5 = 2 left over. ☐

Q4 AND is true only when both inputs are true. One valid ticket or card alone is not enough. OR needs at least one true input; NOT reverses a value. ☐

Code First - Model

```
characters = 4
bytesPerCharacter = 2
sizeBytes = characters * bytesPerCharacter
print(sizeBytes)
```

Output: 8

We Do - Check the model

1. Do not claim all interpreters execute exactly one source line directly. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 More readable and often easier to develop/maintain, or portable with suitable runtimes. ☐

Q3 Compiler translates a program to executable/intermediate output; interpreter executes through an interpreting environment. ☐

Q4 Instructions use a machine-specific instruction set understood by that CPU. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: